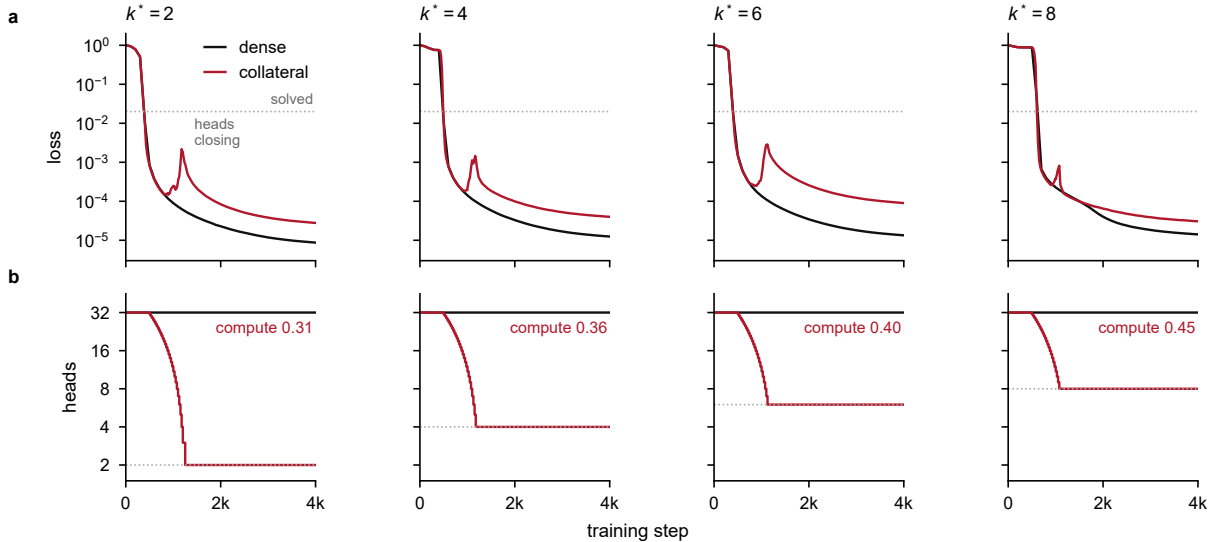




c

Method	Head score	During training	Gradual fade	Reopens	Gate learned	exact count never broke
Collateral rule (ours)	$\min_W L_h - \min_W L$ Optimal Brain Surgeon (Hassibi & Stork 1993)	yes	yes	yes	no	29/30 30/30
<i>Published methods</i>						
ZipLM-style (Kurtic et al. 2023)	$\min_W L_h - \min_W L$ Optimal Brain Surgeon (Hassibi & Stork 1993)	no	no	no	no	24/30 0/30
Michel et al. 2019	$ \partial L / \partial g_h $ gradient of the head gate	yes	no	no	no	28/30 0/30
Learned gates (Voita et al. 2019)	$L + \lambda \sum_h P(g_h \neq 0)$ L0 penalty, best of 3	yes	no	no	yes	9/30 0/30
<i>Ablations of our rule (one part removed)</i>						
no fade (instant closing)	$\min_W L_h - \min_W L$ Optimal Brain Surgeon (Hassibi & Stork 1993)	yes	no	yes	no	30/30 0/30
no re-fit	$L_h(W) - L(W)$ Optimal Brain Damage (LeCun et al. 1990)	yes	yes	yes	no	2/30 30/30
no score (random choice)	random head	yes	yes	yes	no	30/30 6/30

Tasks needing 2, 4 and 8 heads, 10 seeds each.

